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Title: Application of Foundation Model and Agile Modeling to Passive Acoustic Detection of Orcas in Monterey Bay National Marine Sanctuary

Monterey Bay Marine National Sanctuary (MBNMS) is an important habitat for three ecotypes of killer whales (orcas, *Orcinus orca*). Transient (Bigg's) orcas prey on gray whale calves and other mammals in most years. Endangered Southern Resident orcas, which prey on fish, have been seen in the region in multiple years. On occasion a pod of Offshore orcas, whose preferred prey are sharks, is observed in the region. While seen in MBNMS, none of the ecotypes are yet routinely detected and classified from passive acoustic monitoring (PAM) data. Continuous PAM from the center of MBNMS, through the Monterey Accelerated Research System (MARS), located at Smooth Ridge (36°42.75' N, 122°11.21' W) in Monterey Bay 30 km from shore and at about 900 m depth, provides the opportunity to advance knowledge of orcas in the region. MARS hosts a permanently deployed Ocean Sonics icListen HF digital broadband omnidirectional hydrophone (frequency range of 10 Hz to 100 kHz), whose signals are relayed back to shore in real time, 24/7 using the cabled ocean observatory. Since acoustic detections are rare, and manual annotation of sound recordings tedious, acoustic machine learning training data for species or ecotype detection and classification are hard to come by. We attempt to apply pre-trained community released machine learning models to the thousands of hours of recorded data to gain insights into the distribution of orcas in this habitat over time.

Application of Google multi-species whale CNN model and OrcAI CNN model to resampled Pacific Sound MARS recordings yielded high confidence detection of orca calls in Monterey Bay. The Google multi-species whale model was trained by its authors on hundreds of recorded vocalizations of seven classes of whales (including orcas), and five vocalization variations. The OrcAI model was trained by its authors on hours of acoustic signals from Icelandic fish-eating orca recordings, with pulsed calls dominating the training signals. However, each model produced significant confusion in our Pacific Sound MARS recordings between orca calls, Pacific white-sided Dolphin calls (*Lagenorhynchus obliquidens*), and other detections, reducing the usefulness of these models to identify times that orcas were present in Monterey Bay.

Here we report the results of our efforts to apply Perch 2.0 CNN foundation model to Pacific Sound archive recordings, using a small collection of labeled spectrograms of orca calls, dolphin calls, and other signals, and applying active learning (agile modeling) with a vector database in a system called Perch-Hoplite. Perch 2.0 is a pre-trained model for bioacoustics, trained in a supervised manner for off-the-shelf classification of thousands of multi-taxa vocalizations as well as strong embeddings for transfer learning. The model uses a convolutional architecture trained on mel-scaled log-spectrograms, along with a form of self-distillation and self-supervised loss with a goal of producing strong embeddings that are linearly separable to enable use in a range of bioacoustics tasks. Hoplite is a system for storing large volumes of embeddings to use for vector search and active learning workflow. We apply a small number of labeled spectrograms to transfer on fixed embeddings to enable one-shot retrieval through embedding vector search.

We compare our acoustic detections with visual sightings documented by the California Killer Whale Project and reports from vessels in Monterey Bay.

This work shows to the OCEANS community how a bioacoustics foundation model performs in enabling insights into marine mammal populations, and how it can be used to produce a custom classifier with a small amount of labeled data, which is efficient in researchers' time and in computational resources, and supports ecological and population assessment and protection.

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a short abstract/summary (portal entry for indexing and initial processing, max 2250 characters) :

Monterey Bay National Marine Sanctuary is an important habitat for three ecotypes of killer whales (*Orcinus orca*): Transient (Bigg's), endangered Southern Resident, and Offshore orcas. Routine automated detection of orcas is being developed for continuous passive acoustic monitoring via a broadband hydrophone deployed at ~900 m depth on the MARS cabled observatory. Manual annotation of the resulting thousands of hours of recordings is impractical, and acoustic training data from the MARS hydrophone for these ecotypes are scarce.

We evaluated two existing CNN models -- the Google multi-species whale model and OrcAI -- against Pacific Sound MARS recordings. While both yielded high-confidence orca detections, both produced significant confusion between orca calls, Pacific white-sided dolphin calls (*Lagenorhynchus obliquidens*), and other signals, limiting their operational utility.

Here we report the application of the Perch 2.0 bioacoustics foundation model with agile modeling (active learning) via the Perch-Hoplite vector database system. Using a small set of labeled audio files of orca calls, dolphin calls, and other signals, we perform transfer learning on fixed embeddings, enabling efficient one-shot retrieval through vector search. Acoustic detections are compared against visual sighting records from the California Killer Whale Project and others.

This work demonstrates how a bioacoustics foundation model, combined with a small labeled dataset and active learning, can produce a custom classifier efficiently -- requiring minimal researcher time and computational resources -- while supporting ecological assessment and protection of marine mammal populations.

Appendix (not for abstract)

Maybe also try public orca data

[A Public Dataset of Annotated Orcinus orca Acoustic Signals for Detection and Ecotype Classification | Scientific Data](#)

ETP Eastern Tropical Pacific orcas also seen in Monterey Bay. Distinctive vocalizations
<https://www.tiktok.com/@orcasonewild/video/7605440316593769736>

Orcas that hunt Sperm whale are also seen in Monterey Bay.
<https://onlinelibrary.wiley.com/doi/full/10.1111/mms.12956>